Data Visualization

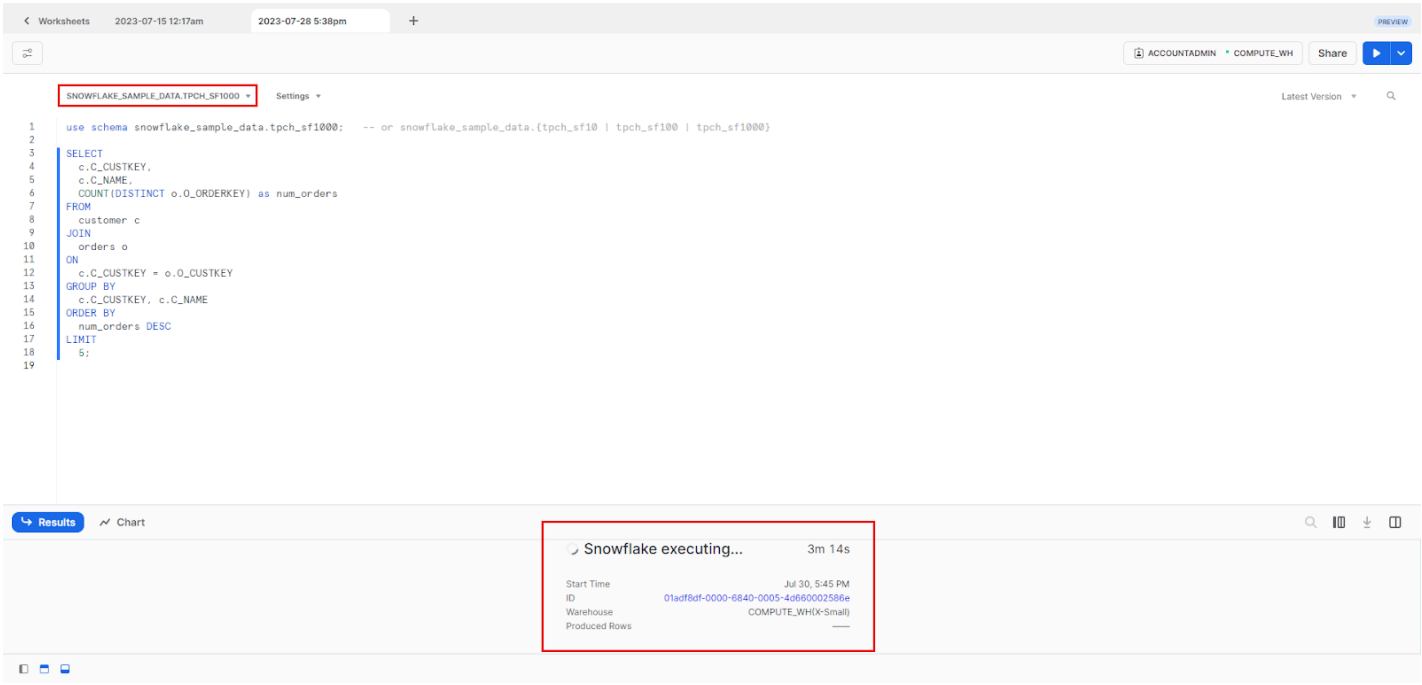
In this step we are going to visualize our queried data in Tableau using Snowflake Custom Query in Tableau. We are going to try different graphs and charts.

Step 1

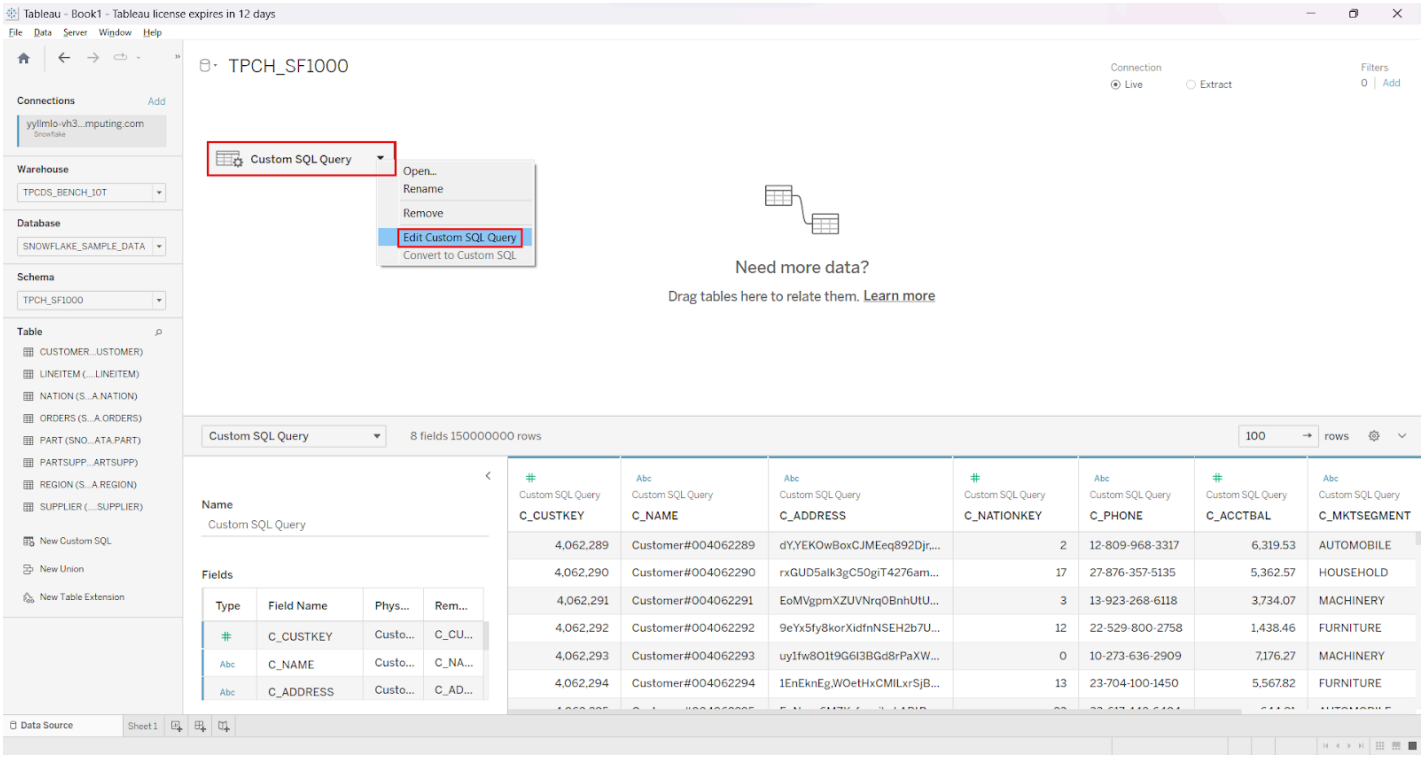
- Go to your snowflake console and then start preparing your queries for visualization. Here is a query to Find the Top 5 Customers with the Highest Number of Orders in Snowflake.

```
SELECT
c.C_CUSTKEY,
c.C_NAME,
COUNT(DISTINCT o.O_ORDERKEY) as num_orders
FROM
customer c
JOIN
orders o
ON
c.C_CUSTKEY = o.O_CUSTKEY
GROUP BY
c.C_CUSTKEY, c.C_NAME
ORDER BY
num_orders DESC
LIMIT  5;
```

Notice the schema that is set using ‘use schema snowflake_sample_data.tpch_sf1000;’. Also check the query after running using play button or Ctrl+Enter on your keyboard, is taking longer time using a small warehouse.

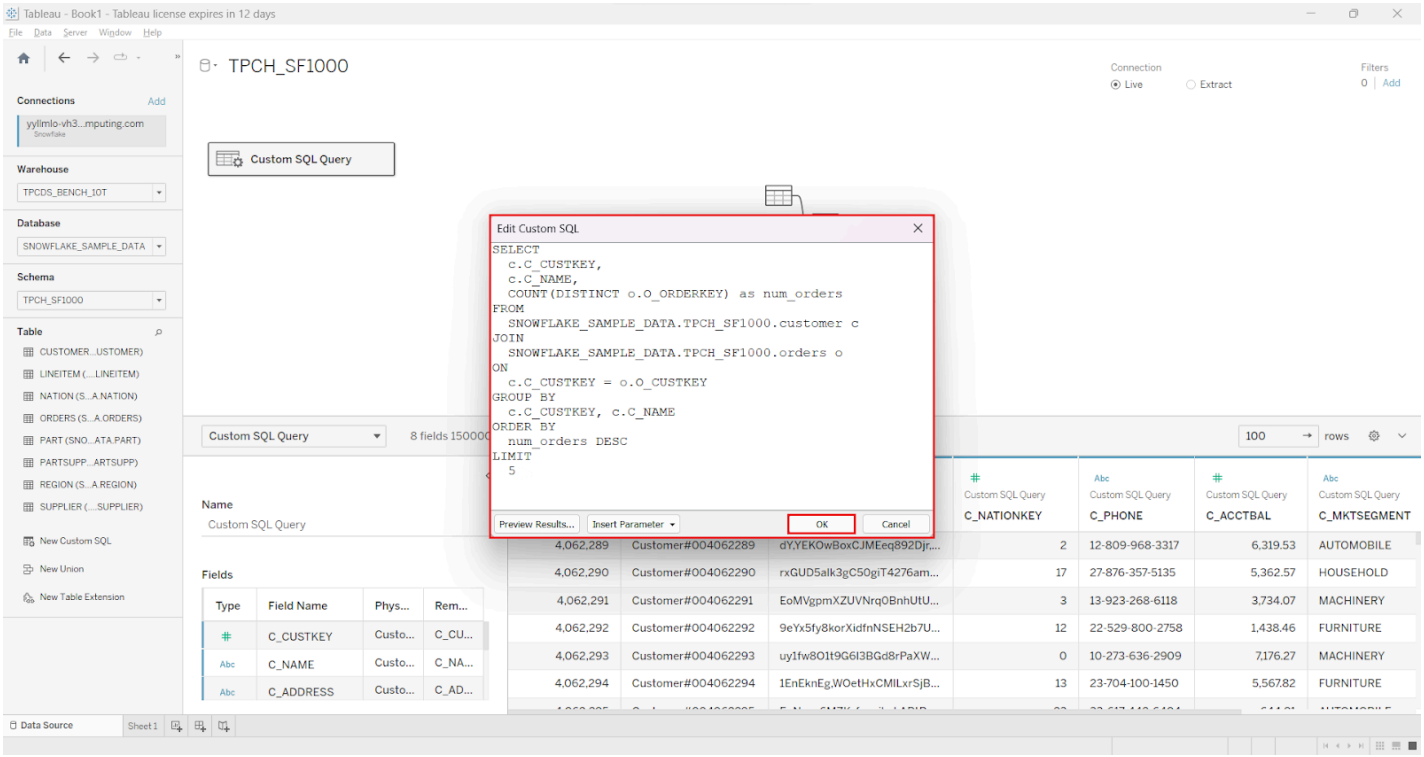


- Try running the same query on Tableau Desktop by click on Custom SQL Query on the data pane and then click on Edit Custom SQL Query.

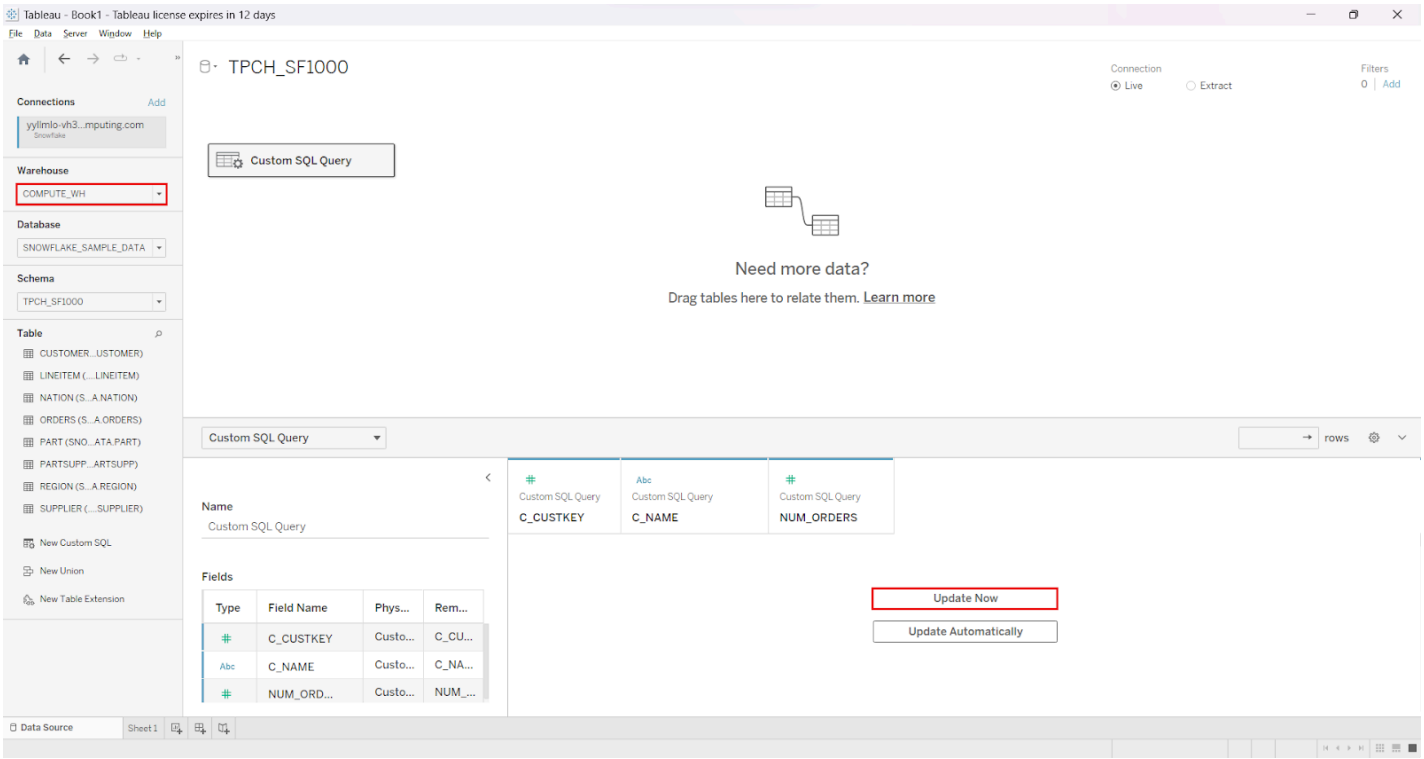


Now paste the below query from snowflake and click on OK.

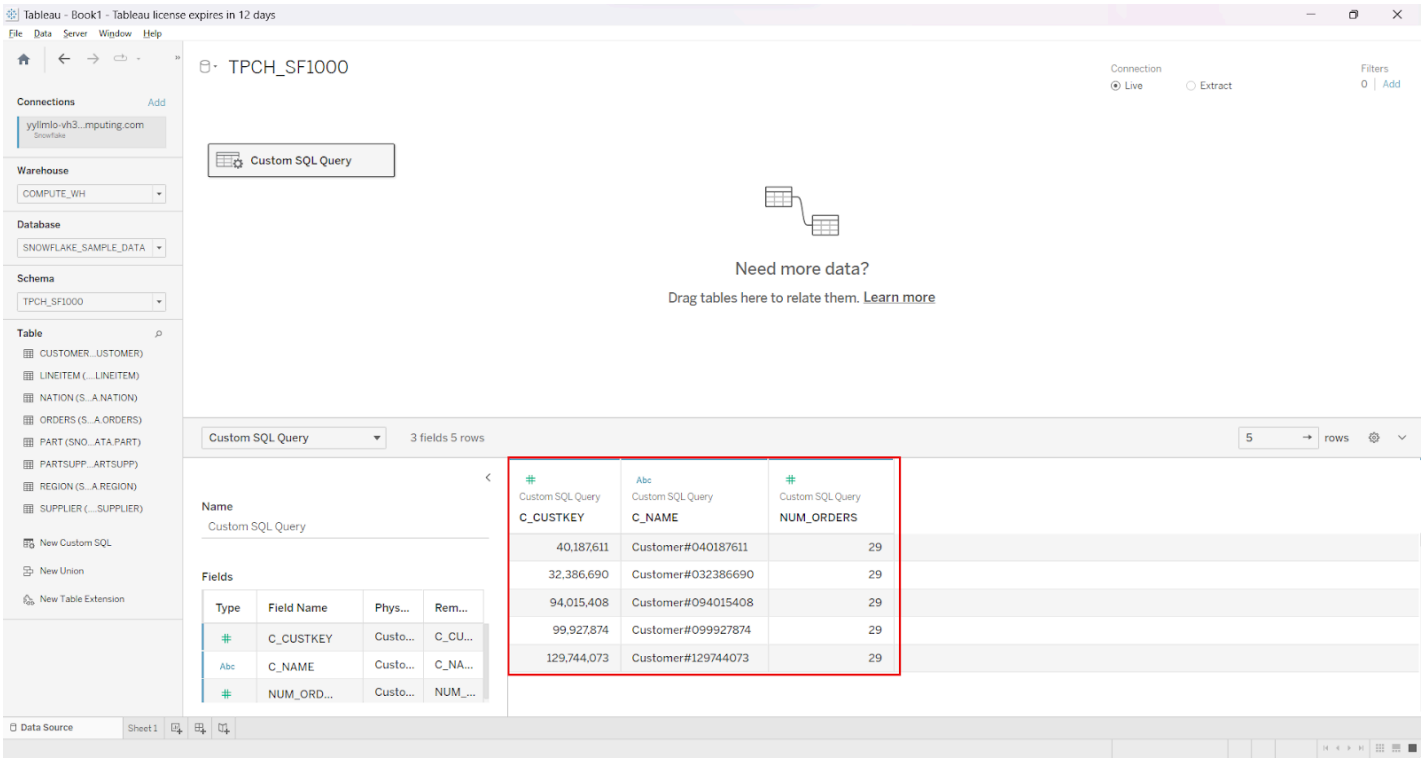
```
SELECT
  c.C_CUSTKEY,
  c.C_NAME,
  COUNT(DISTINCT o.O_ORDERKEY) as num_orders
FROM
  SNOWFLAKE_SAMPLE_DATA.TPCH_SF1000.customer c
JOIN
  SNOWFLAKE_SAMPLE_DATA.TPCH_SF1000.orders o
ON
  c.C_CUSTKEY = o.O_CUSTKEY
GROUP BY
  c.C_CUSTKEY, c.C_NAME
ORDER BY
  num_orders DESC
LIMIT
  5
```



Then click on Update Now to check the tableau computation using same warehouse.



- You will be able to see the data.

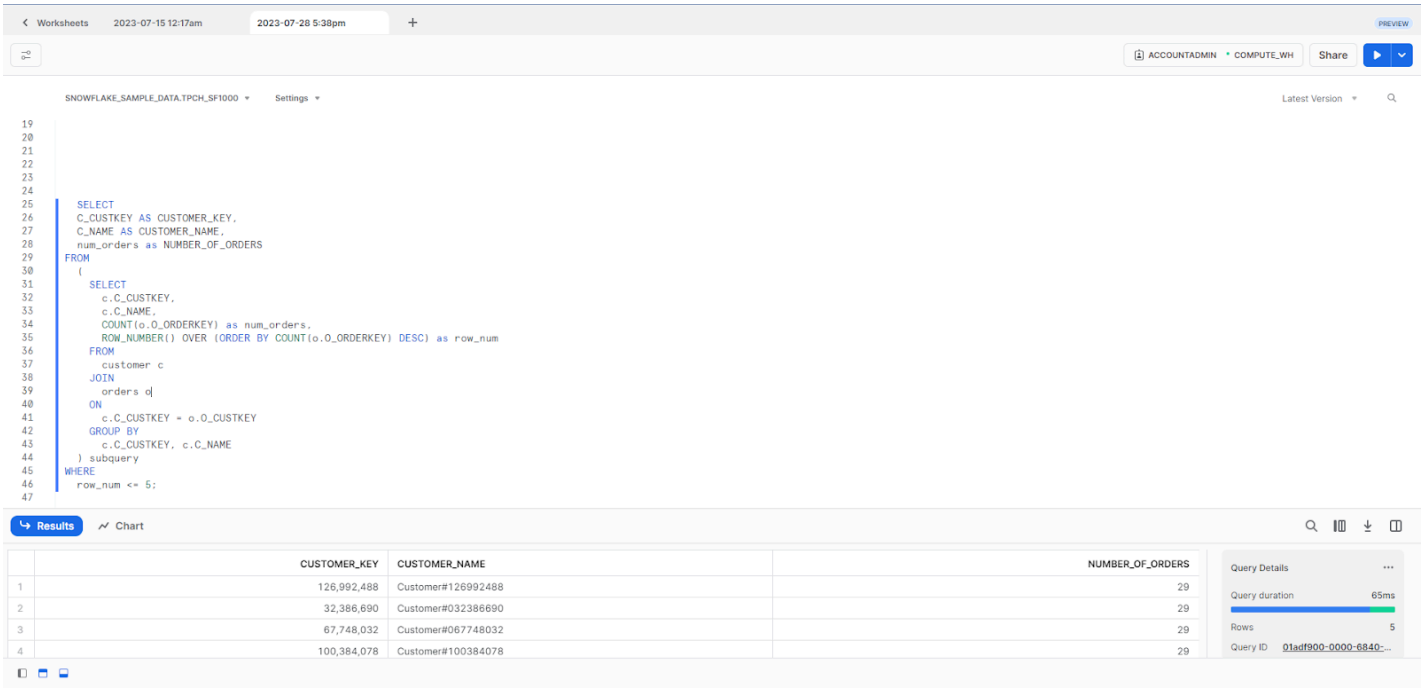


Notice Tableau did take much time to give the result like snowflake.

- So, to follow a common way of querying we will use the most optimized query as given below. The below query gives results with a small number of seconds above 2 minutes for the first time.

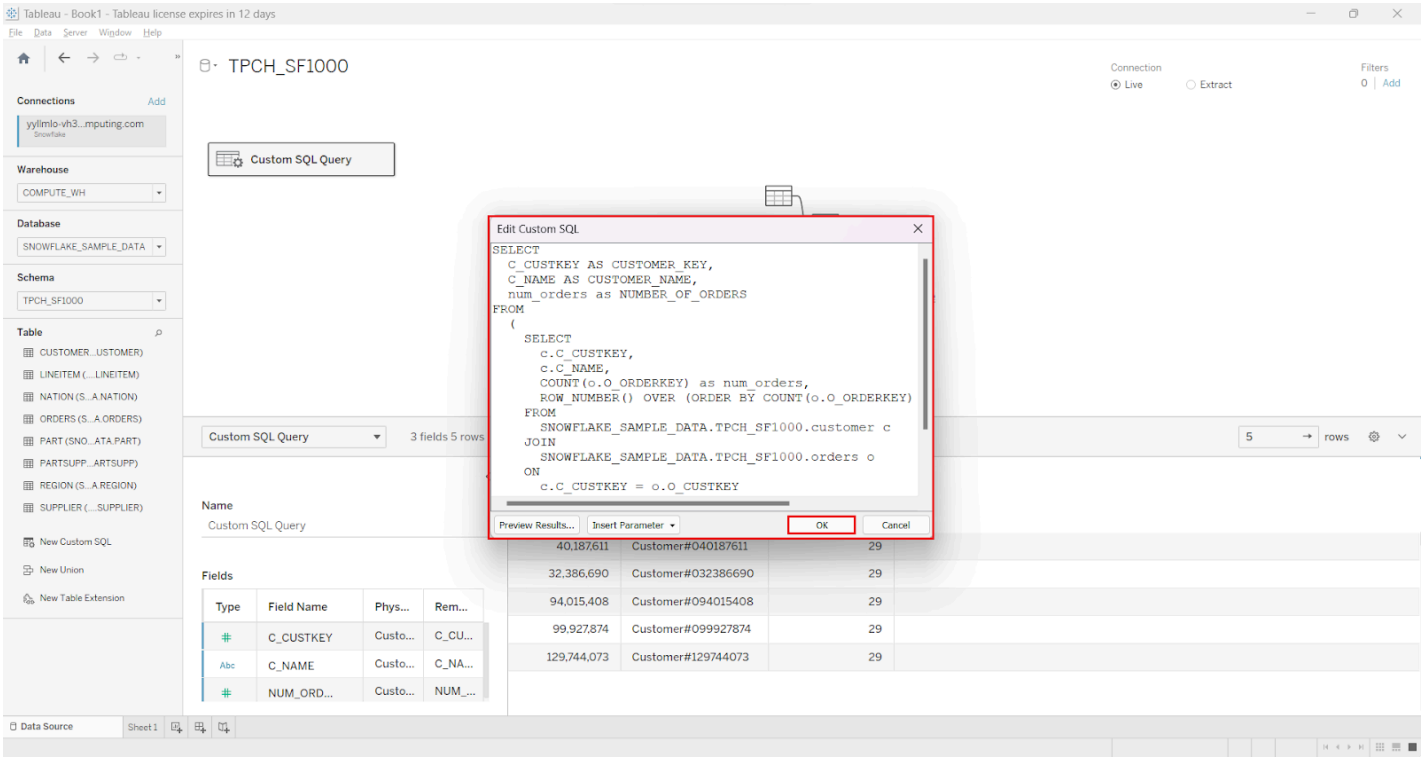
```
SELECT
  C_CUSTKEY AS CUSTOMER_KEY,
  C_NAME AS CUSTOMER_NAME,
  num_orders as NUMBER_OF_ORDERS
FROM
```

```
  (
    SELECT
      c.C_CUSTKEY,
      c.C_NAME,
      COUNT(o.O_ORDERKEY) as num_orders,
      ROW_NUMBER() OVER (ORDER BY COUNT(o.O_ORDERKEY) DESC) as row_num
    FROM
      SNOWFLAKE_SAMPLE_DATA.TPCH_SF1000.customer c
    JOIN
      SNOWFLAKE_SAMPLE_DATA.TPCH_SF1000.orders o
    ON
      c.C_CUSTKEY = o.O_CUSTKEY
    GROUP BY
      c.C_CUSTKEY, c.C_NAME
  ) subquery
WHERE
  row_num <= 5;
```

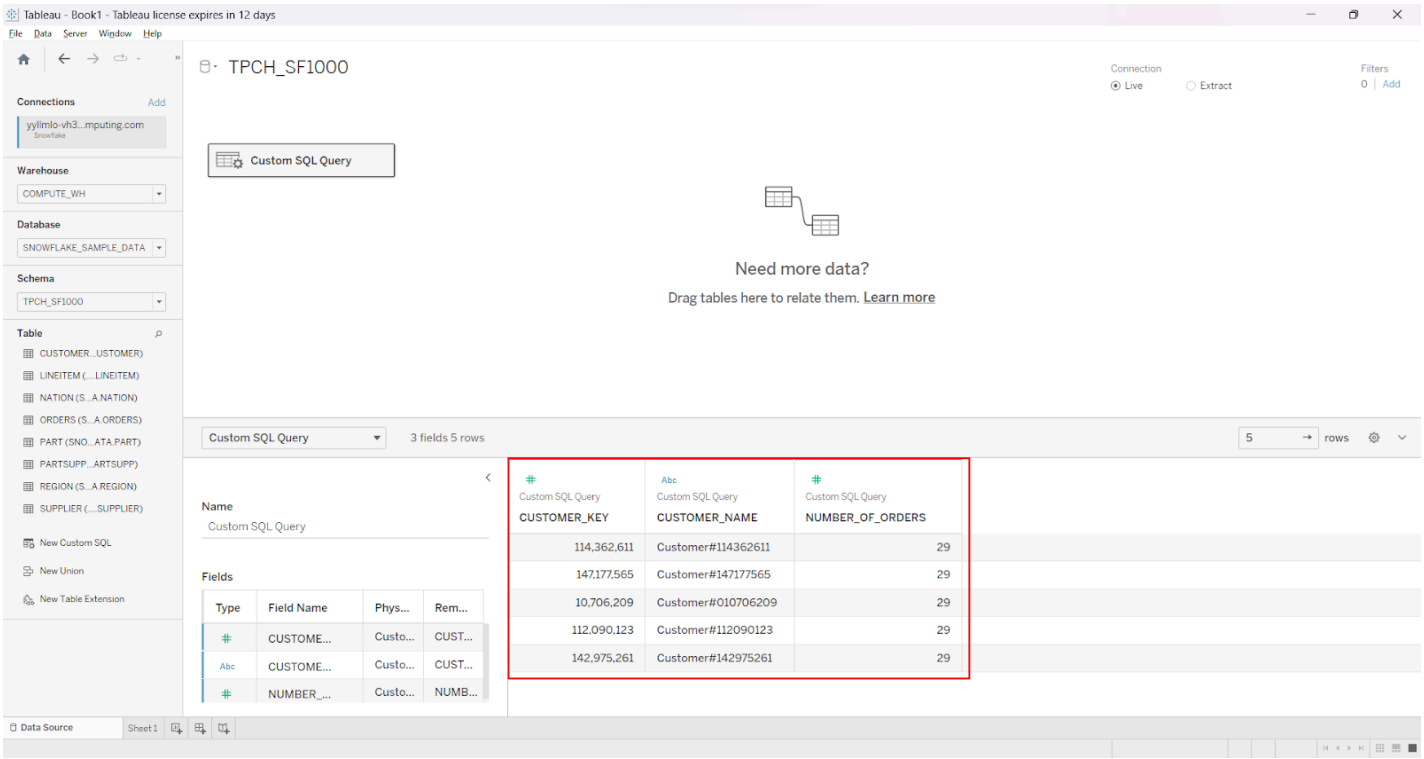


- Let us use the same query on Tableau Desktop. Follow similar step to edit the custom query and then click on update now.

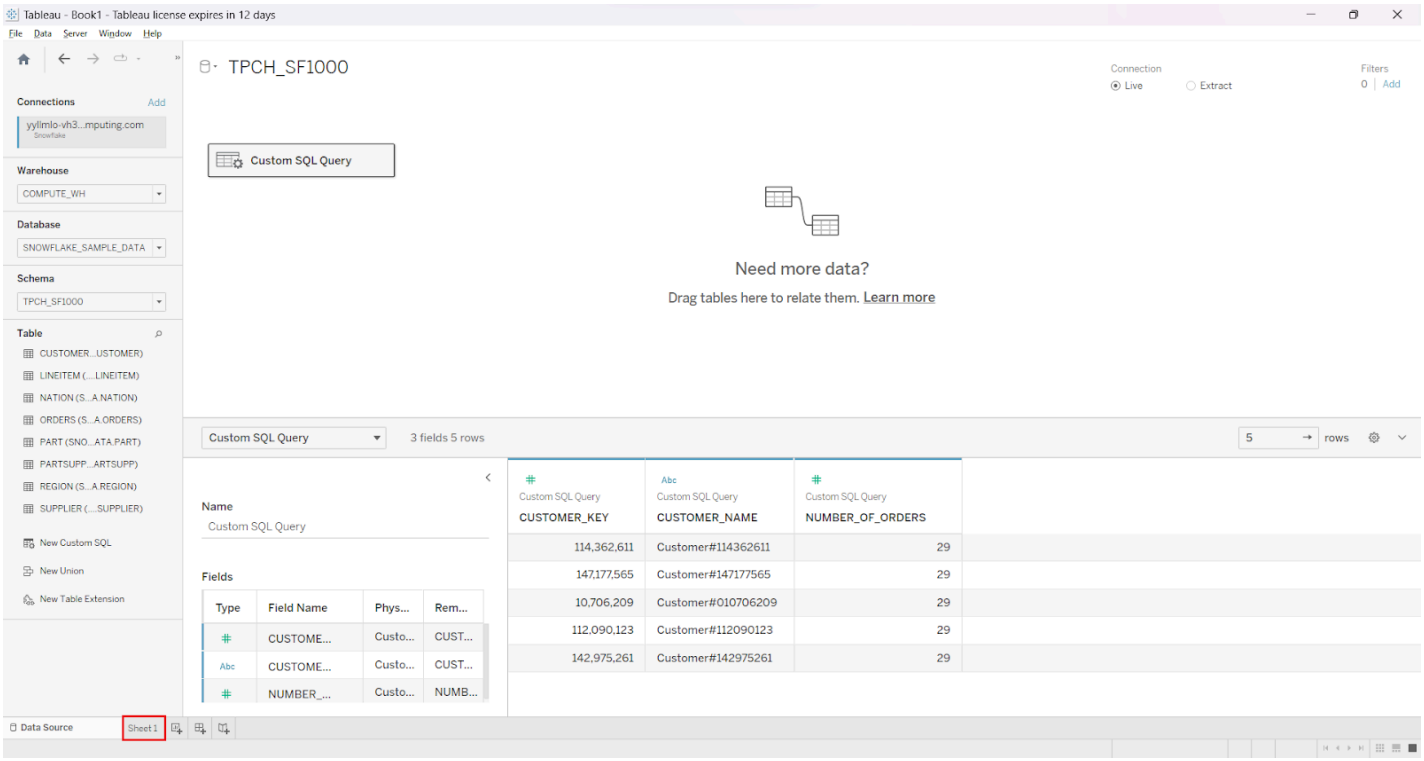
```
SELECT
  C_CUSTKEY AS CUSTOMER_KEY,
  C_NAME AS CUSTOMER_NAME,
  num_orders as NUMBER_OF_ORDERS
FROM
  (
    SELECT
      c.C_CUSTKEY,
      c.C_NAME,
      COUNT(o.O_ORDERKEY) as num_orders,
      ROW_NUMBER() OVER (ORDER BY COUNT(o.O_ORDERKEY) DESC) as row_num
    FROM
      SNOWFLAKE_SAMPLE_DATA.TPCH_SF1000.customer c
    JOIN
      SNOWFLAKE_SAMPLE_DATA.TPCH_SF1000.orders o
    ON
      c.C_CUSTKEY = o.O_CUSTKEY
    GROUP BY
      c.C_CUSTKEY, c.C_NAME
  ) subquery
WHERE
  row_num <= 5
```



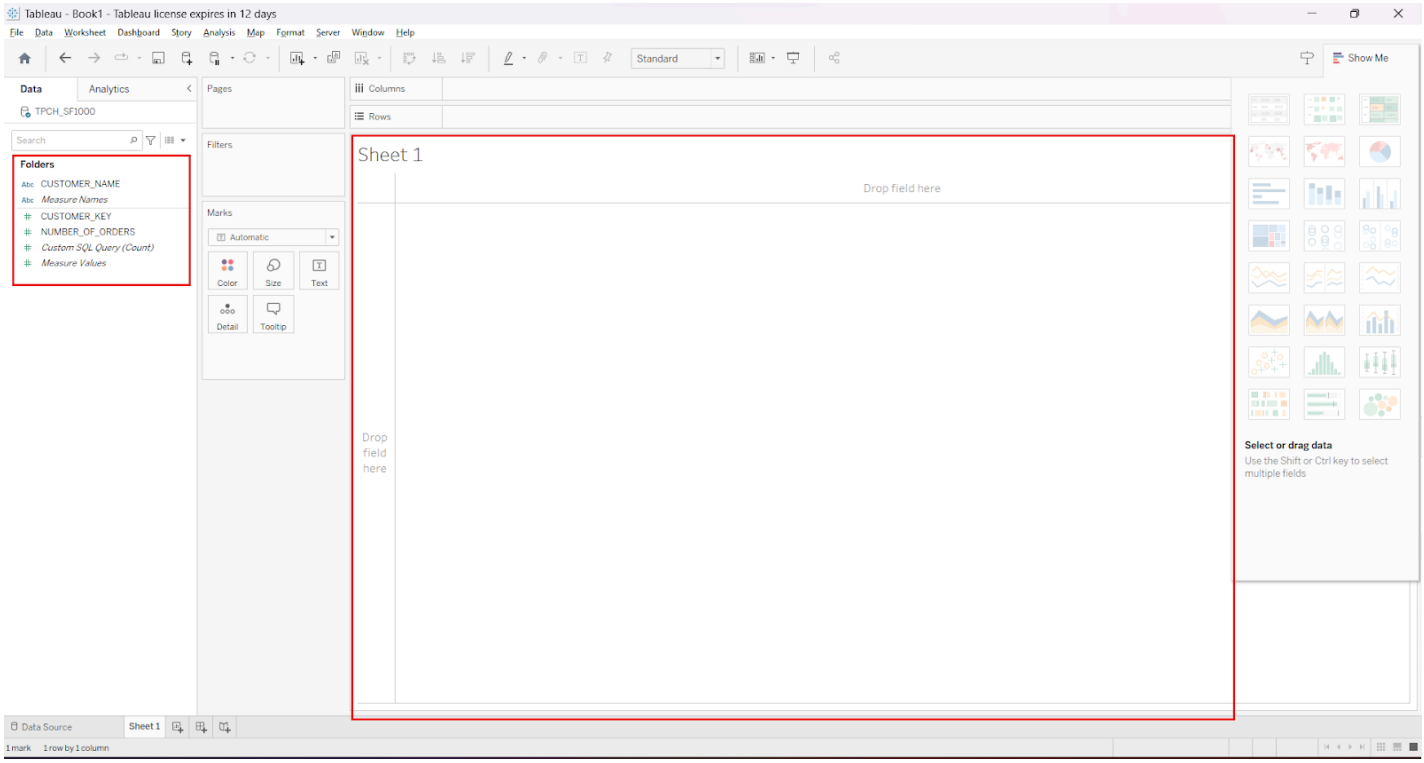
This time it is much faster and the results are on display.



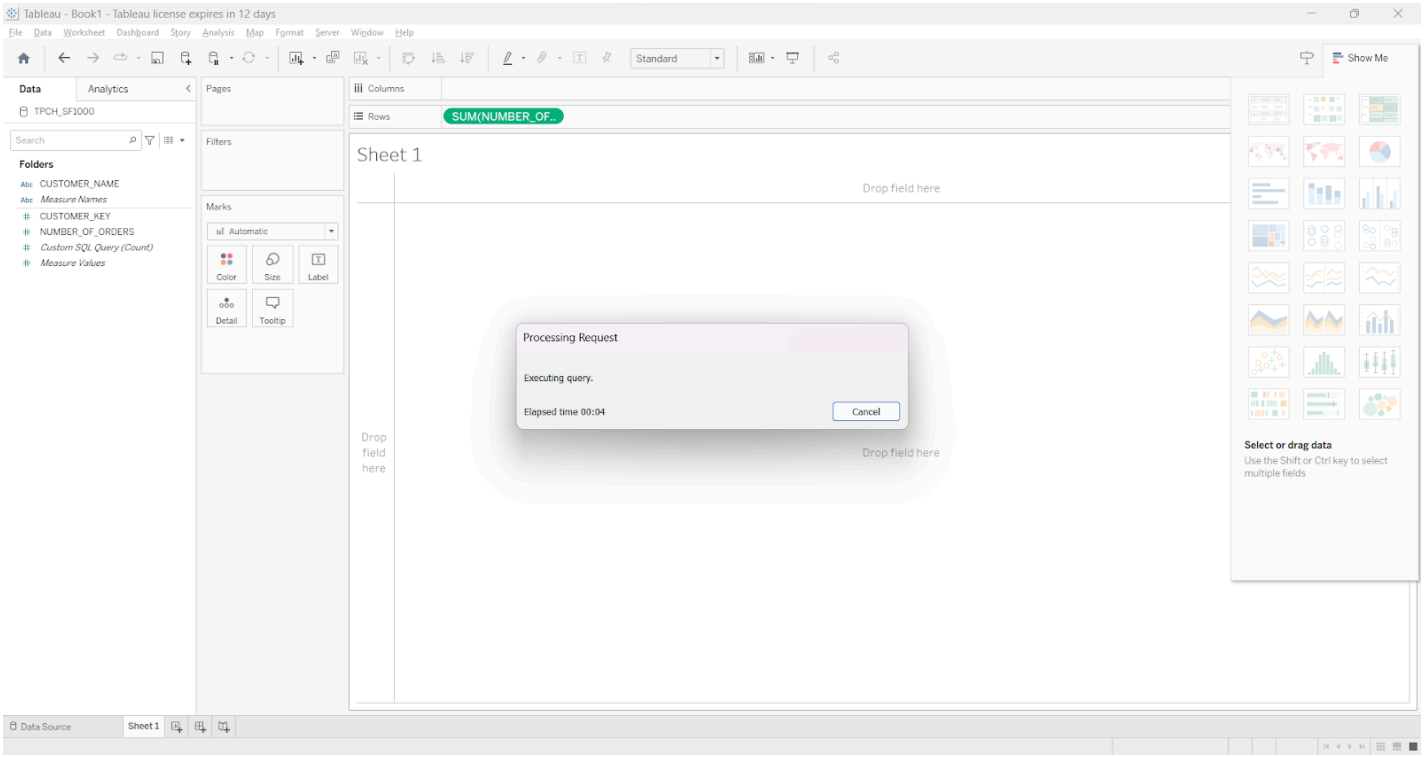
- Now let us go to Sheet1 and prepare for our visualization.



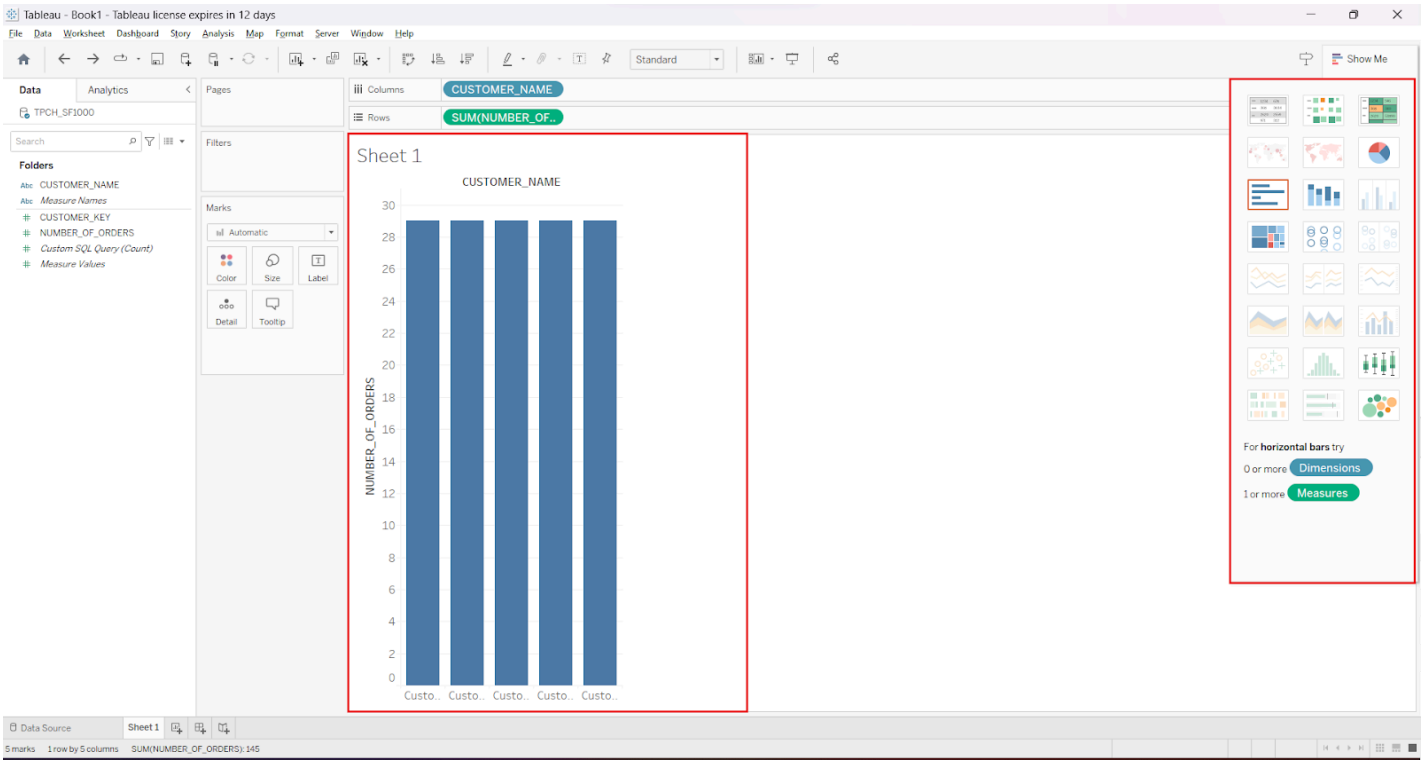
On the next page you will be able to see all the attributes from the query listed. Pick the required dimensions and measures to plot. Here we will plot number of order vs customer key.



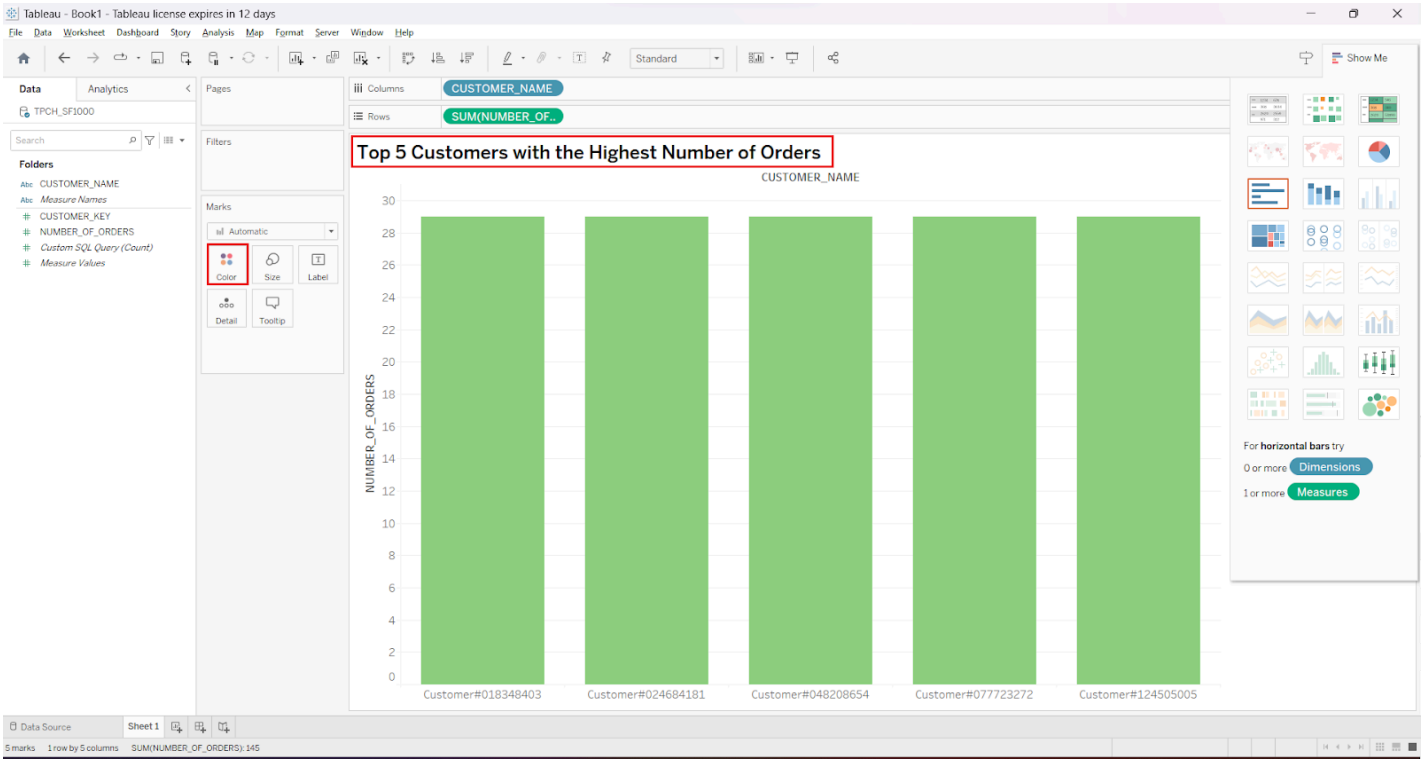
To do it select and drag CUSTOMER_NAME to Columns and Number_Of_Orders to Rows on the plot pane to the right.



- Now you can see a bar graph showing how your data has been queried from the tables. You can adjust the plot by stretching it wide to view better. Notice the available kinds of graphs for you.



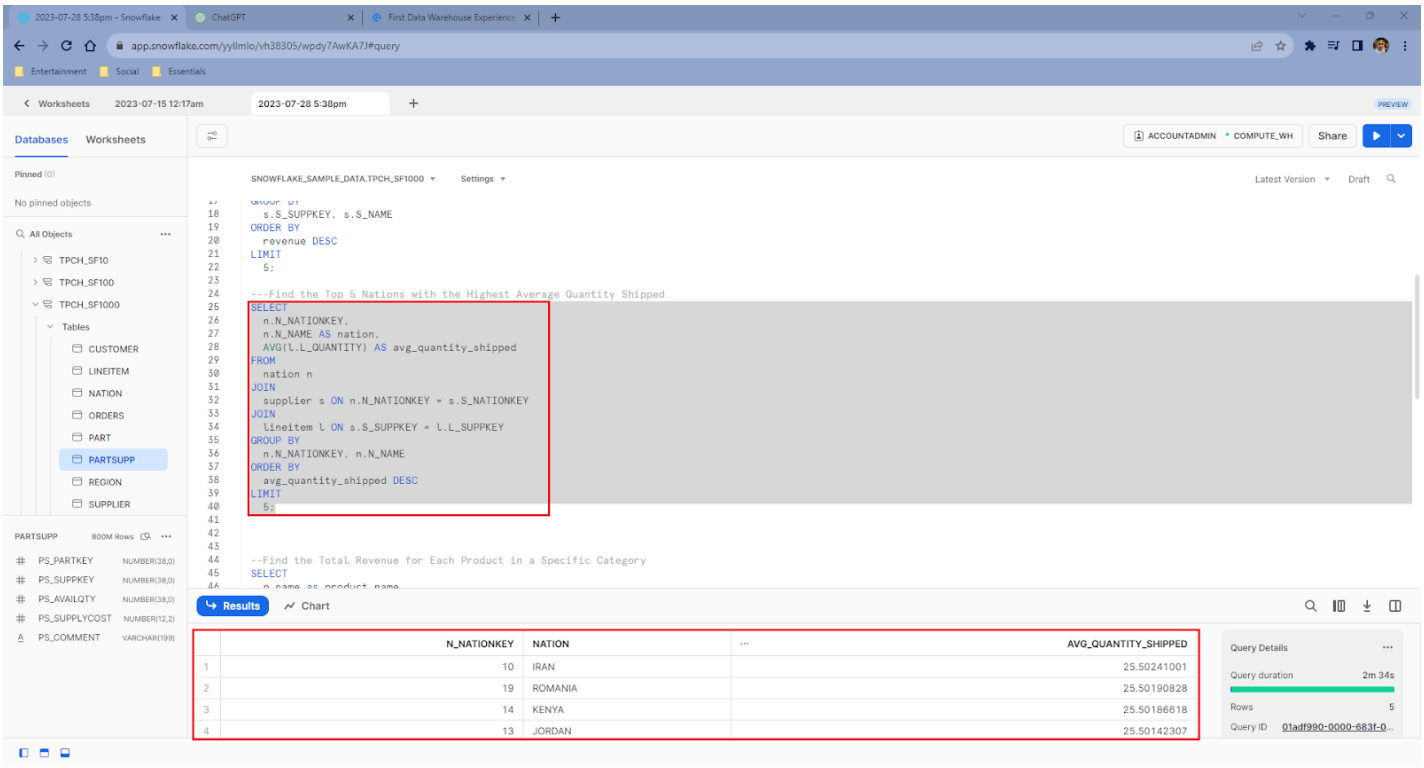
You can also change colours, add labels and many more. After making the plot for better reading here is how it looks. Add a title to your plot.



- Now let us build our next query on Snowflake to Find the Top 5 Nations with the Highest Average Quantity Shipped. Use below query and run in Snowflake.

```
SELECT
  n.N_NATIONKEY,
  n.N_NAME AS nation,
  AVG(l.L_QUANTITY) AS avg_quantity_shipped
FROM
  nation n
JOIN
  supplier s ON n.N_NATIONKEY = s.S_NATIONKEY
JOIN
  lineitem l ON s.S_SUPPKEY = l.L_SUPPKEY
GROUP BY
  n.N_NATIONKEY, n.N_NAME
ORDER BY
```

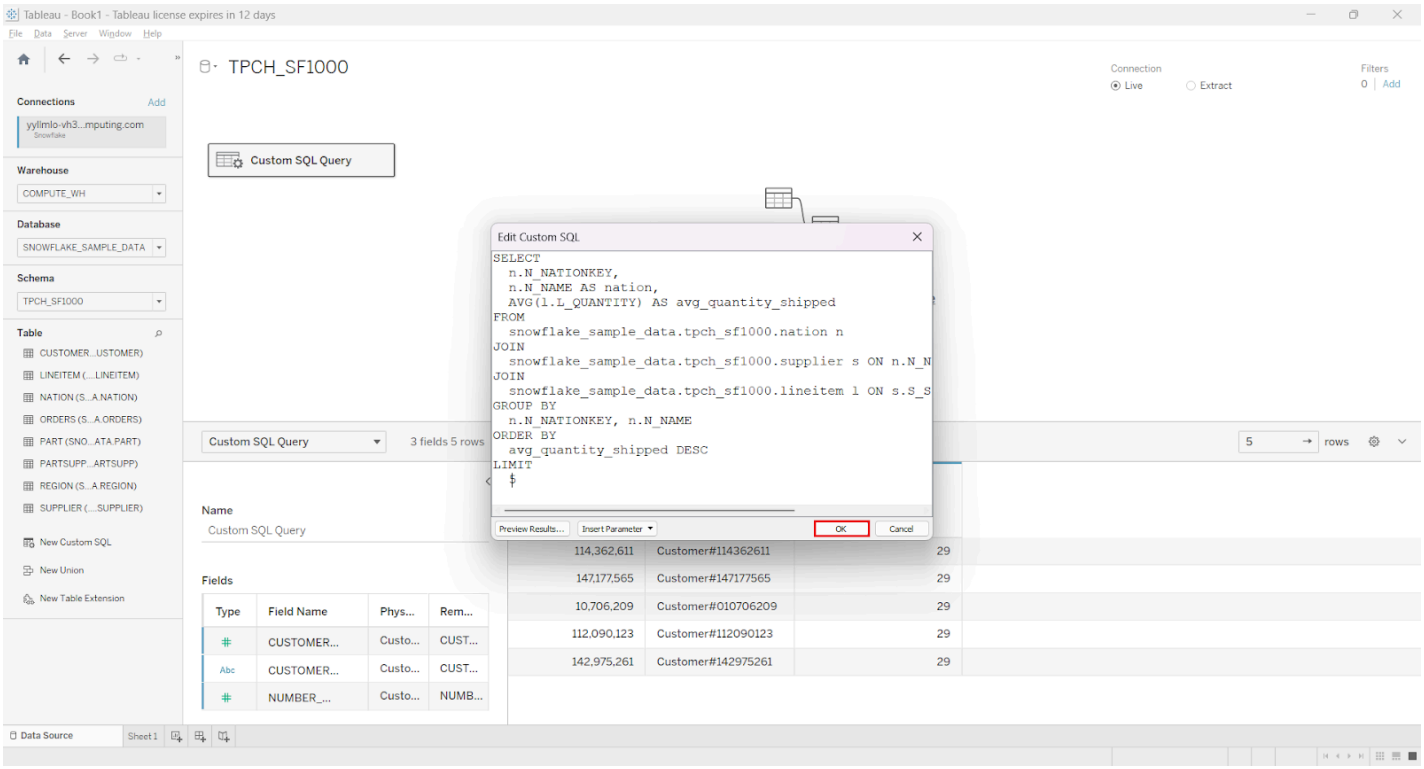
```
  avg_quantity_shipped DESC
LIMIT
  5;
```



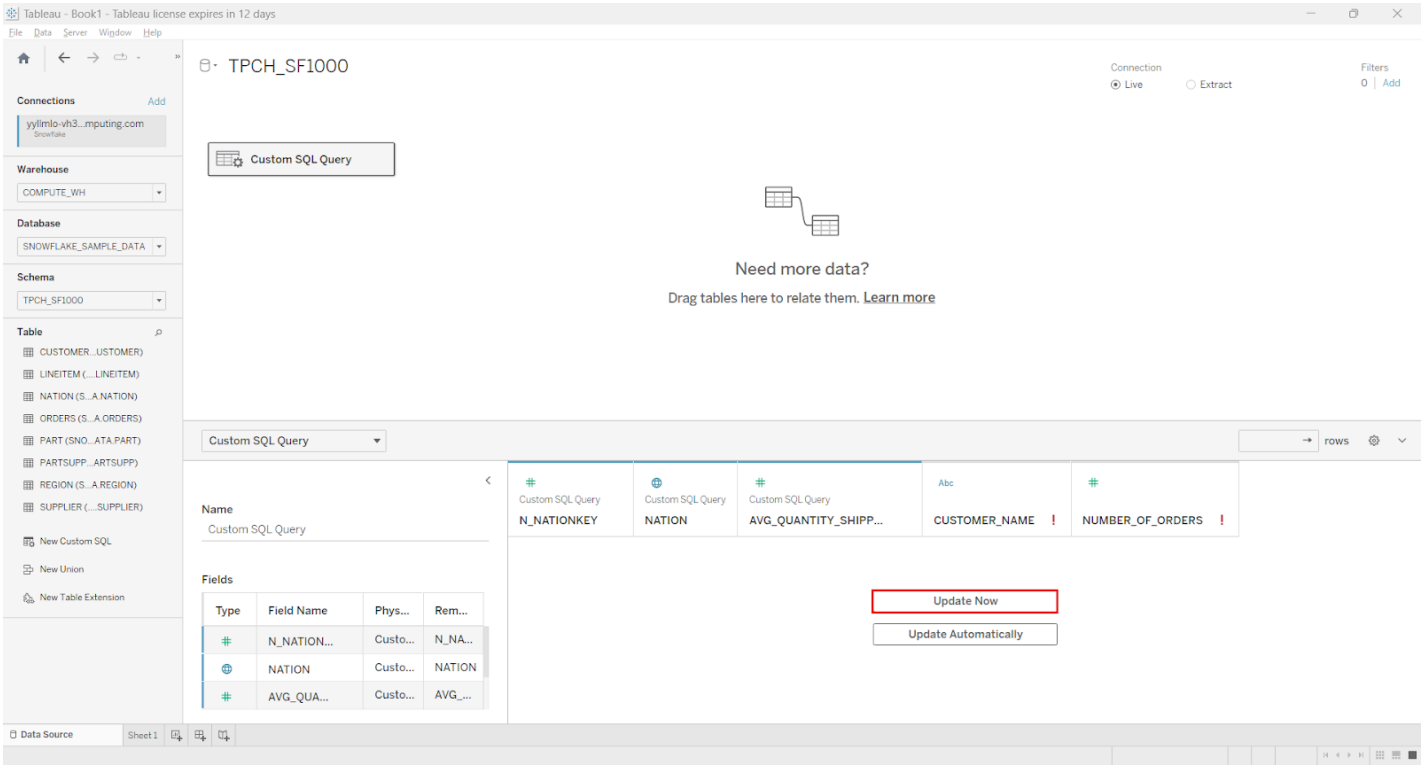
- Let us now edit the custom query in Tableau Desktop and use the new query to fetch the data. Use below query:

```
SELECT
  n.N_NATIONKEY,
  n.N_NAME AS nation,
  AVG(l.L_QUANTITY) AS avg_quantity_shipped
FROM
  snowflake_sample_data.tpch_sf1000.nation n
JOIN
  snowflake_sample_data.tpch_sf1000.supplier s ON n.N_NATIONKEY = s.S_NATIONKEY
JOIN
  snowflake_sample_data.tpch_sf1000.lineitem l ON s.S_SUPPKEY = l.L_SUPPKEY
```

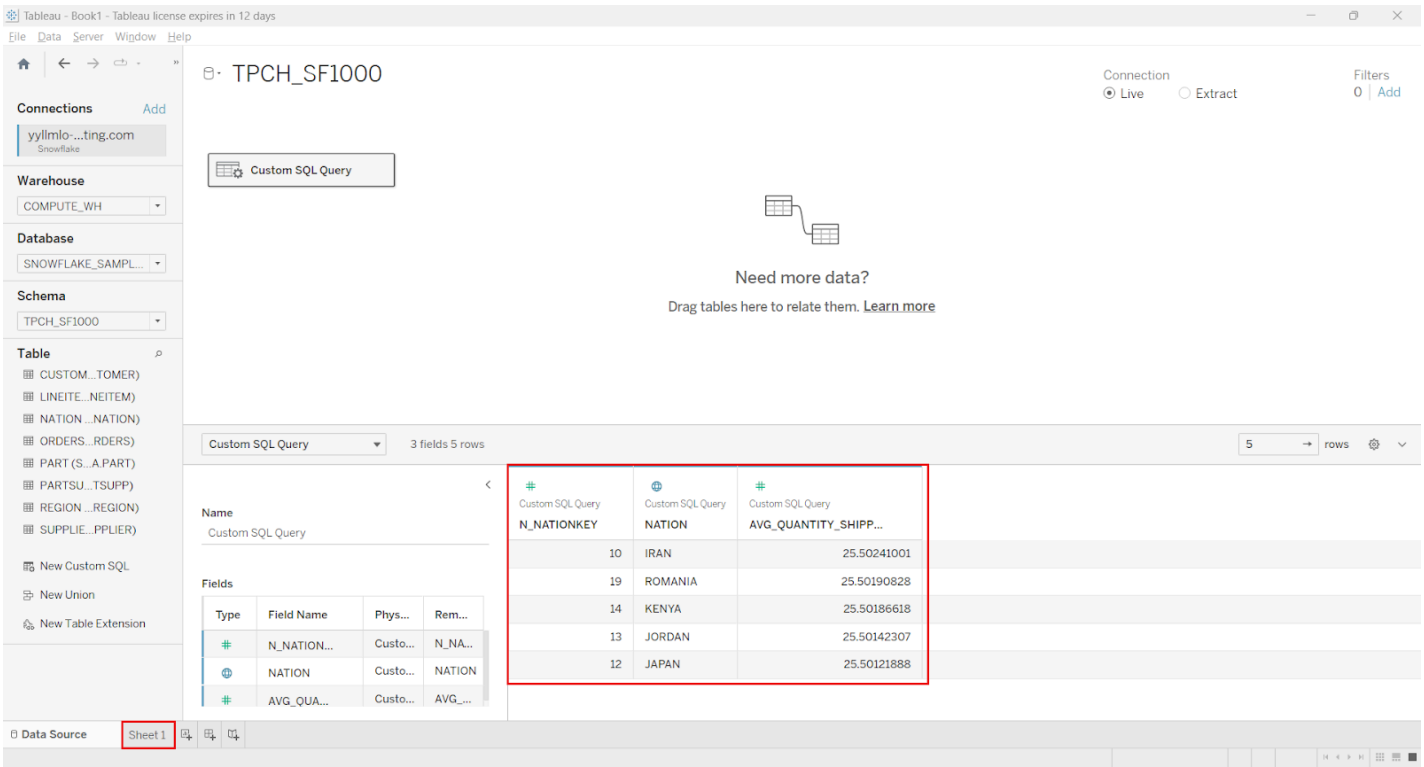
```
GROUP BY
  n.N_NATIONKEY, n.N_NAME
ORDER BY
  avg_quantity_shipped DESC
LIMIT
  5
```



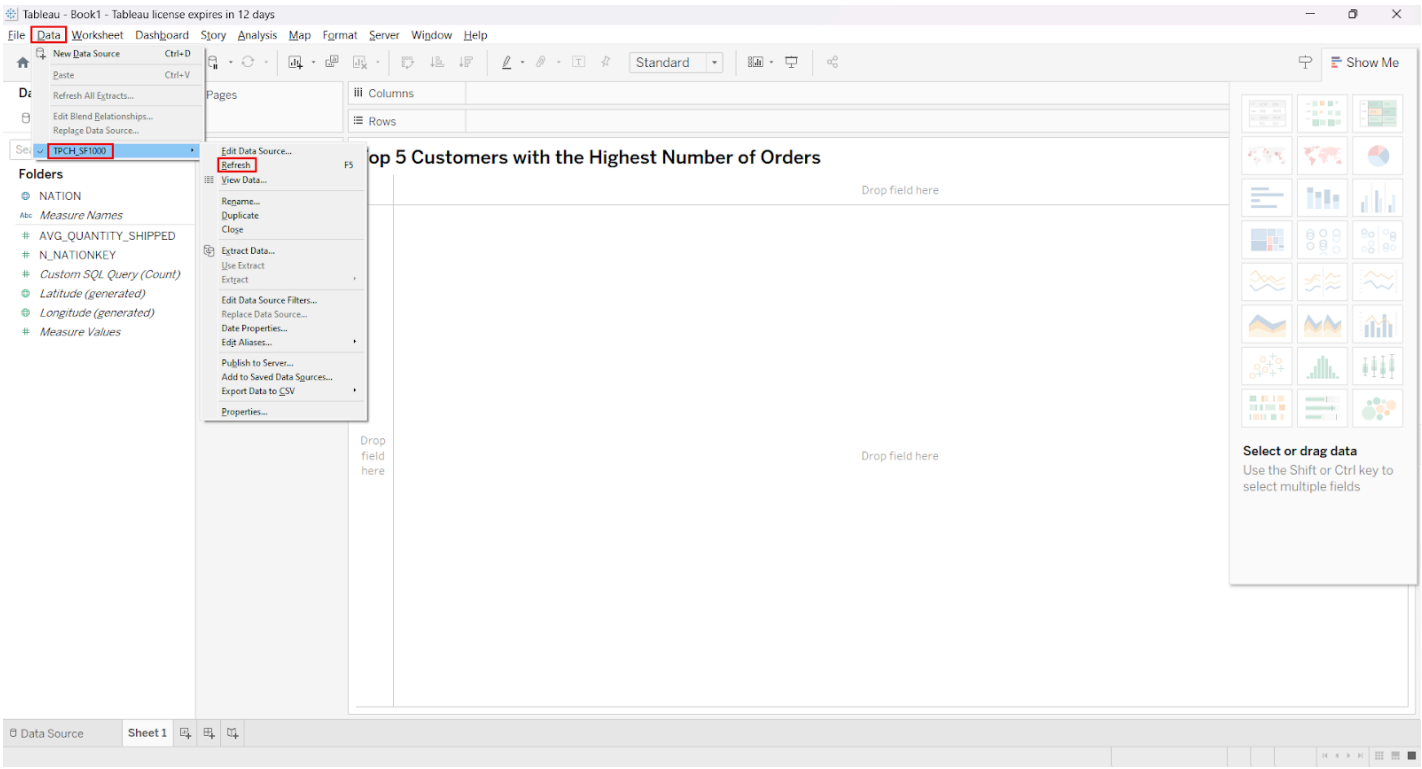
Now update the data in Tableau.



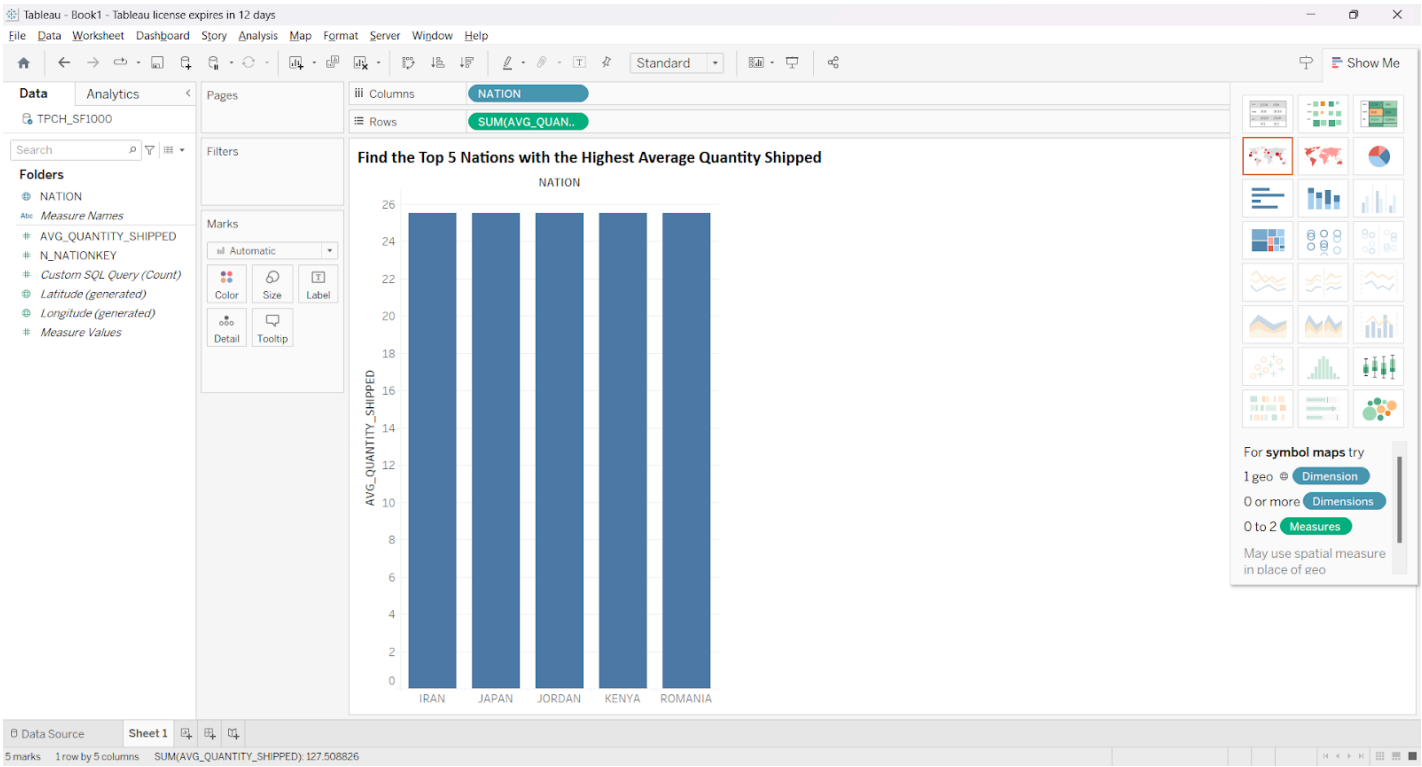
- After the update is done go to sheet1.



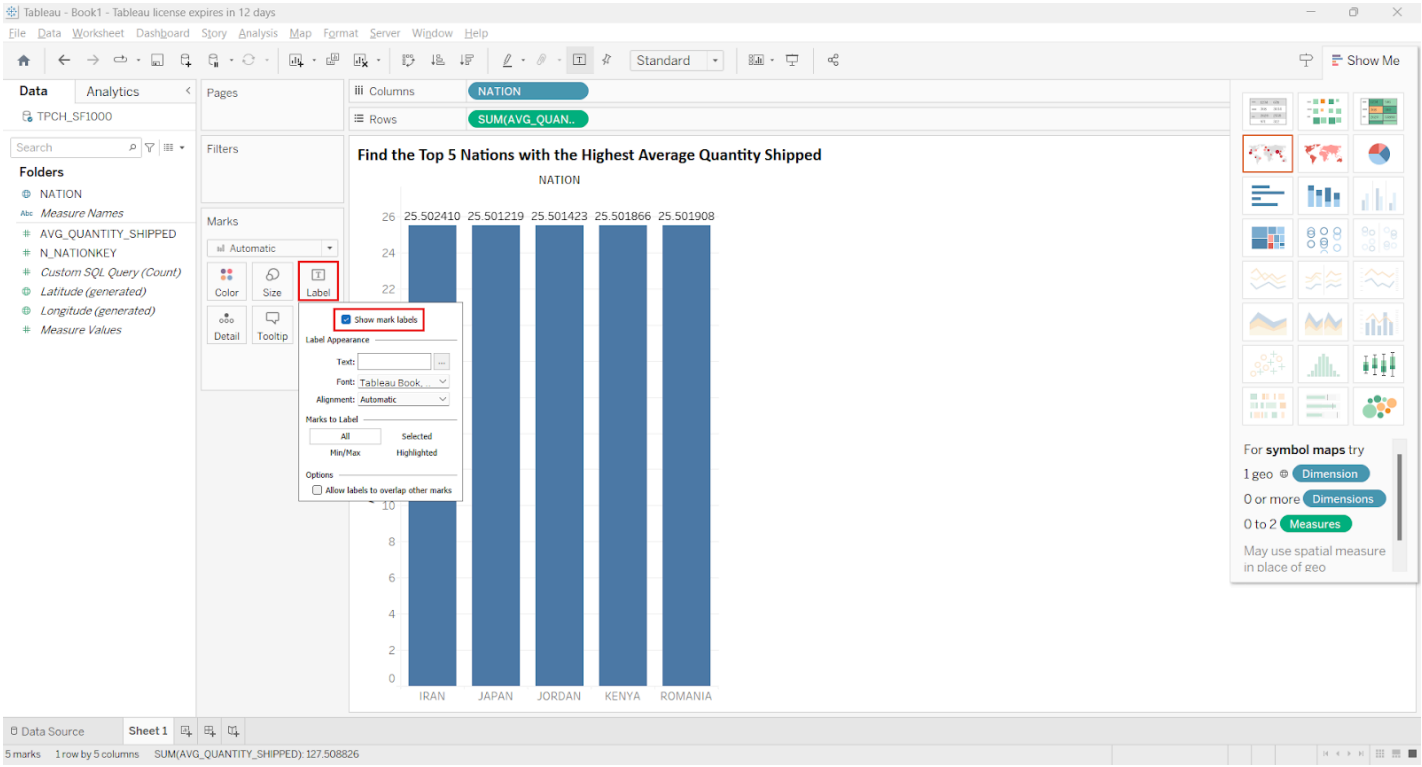
Then click on Data menu then hover over your schema and then click on Refresh data.



- Now you can see the updated dimensions and measures. Now edit the title. And we will plot Nation vs AVG_quantity_shipped. After you have plotted the plot will look like below. You may choose to give different colours and add more labels.



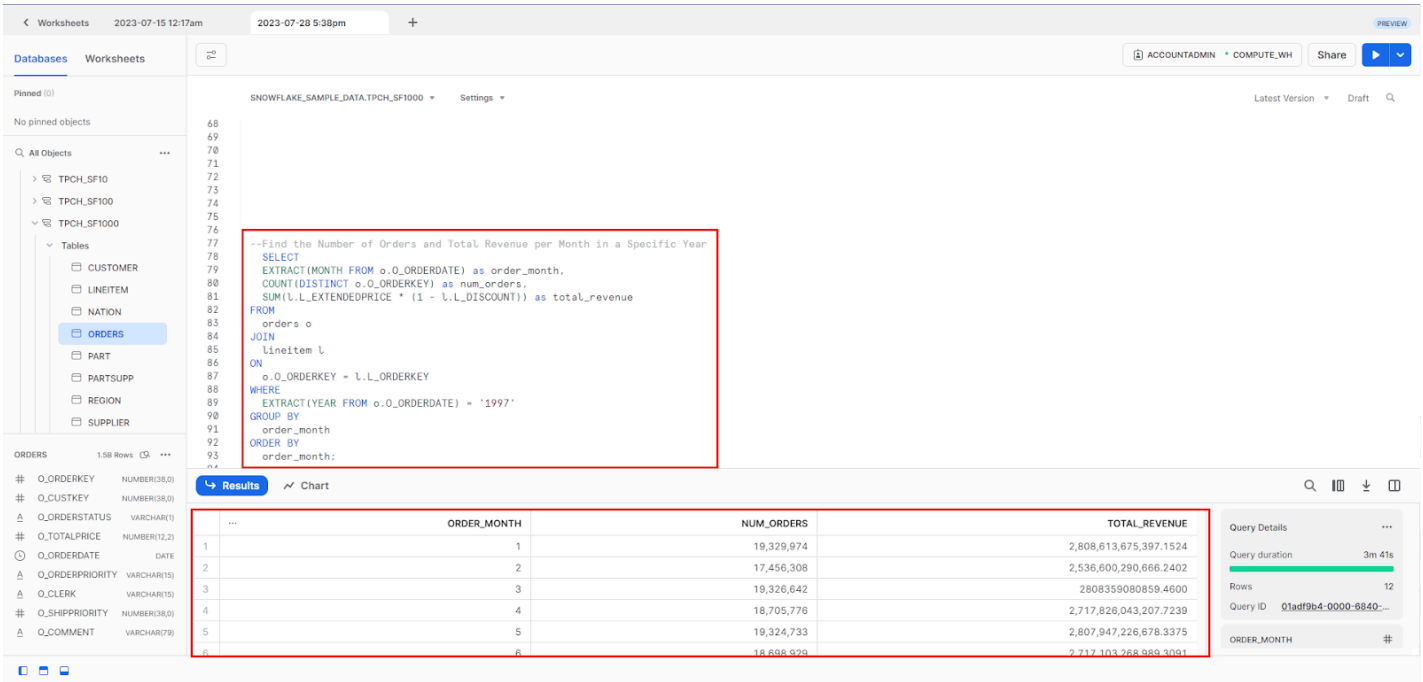
- Now click on Label and then check box on Show mark labels.



- Next let us build our next query on Snowflake to Find the Number of Orders and Total Revenue per Month in 1997. Use below query and run in Snowflake.

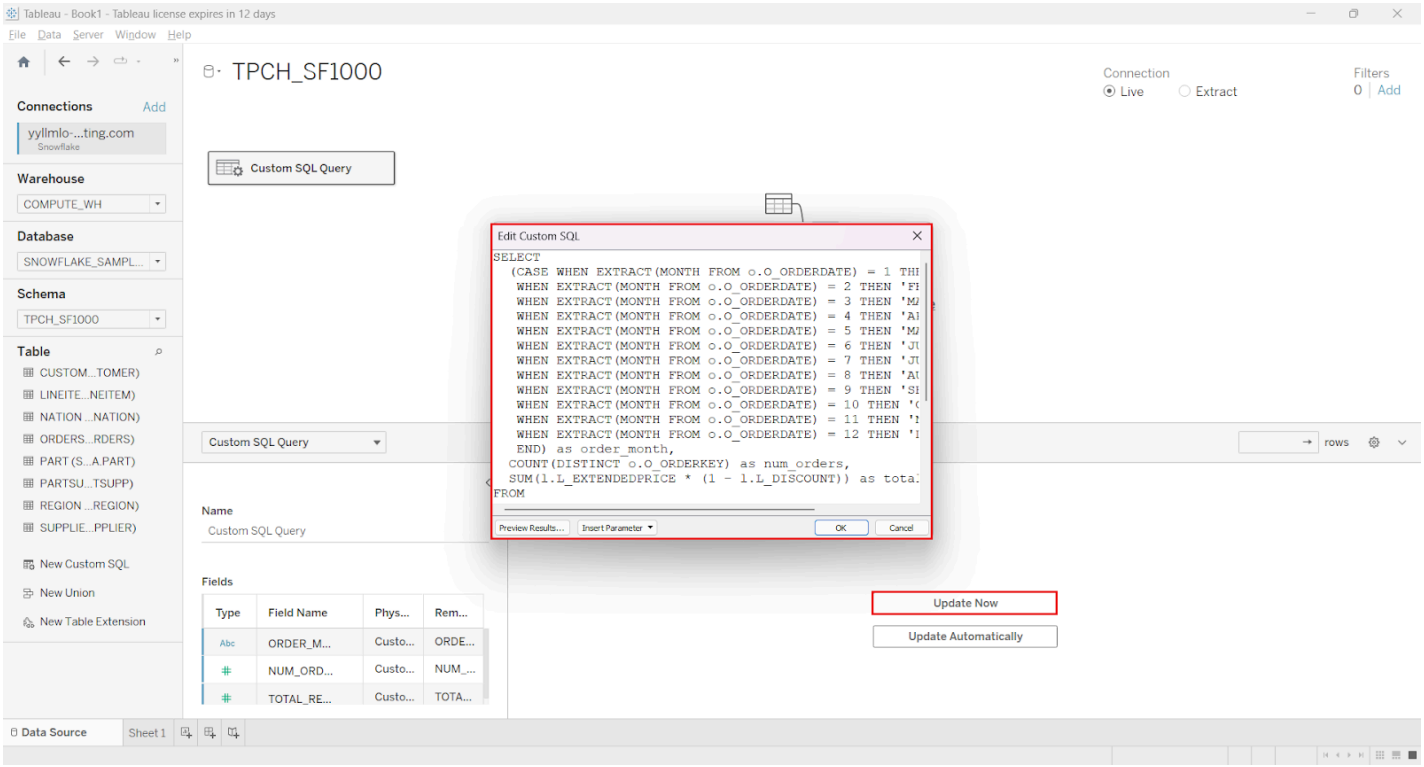
```
SELECT
(CASE WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 1 THEN 'JANUARY'
 WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 2 THEN 'FEBRUARY'
 WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 3 THEN 'MARCH'
 WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 4 THEN 'APRIL'
 WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 5 THEN 'MAY'
 WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 6 THEN 'JUNE'
 WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 7 THEN 'JULY'
 WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 8 THEN 'AUGUST'
 WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 9 THEN 'SEPTEMBER'
 WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 10 THEN 'OCTOBER'
```

```
  WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 11 THEN 'NOVEMBER'
  WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 12 THEN 'DECEMBER'
  END) as order_month,
 COUNT(DISTINCT o.O_ORDERKEY) as num_orders,
 SUM(l.L_EXTENDEDPRICE * (1 - l.L_DISCOUNT)) as total_revenue
FROM
 snowflake_sample_data.tpch_sf1000.orders o
JOIN
 snowflake_sample_data.tpch_sf1000.lineitem l
ON
 o.O_ORDERKEY = l.L_ORDERKEY
WHERE
 EXTRACT(YEAR FROM o.O_ORDERDATE) = '1997'
GROUP BY
 order_month
ORDER BY
 order_month;
```

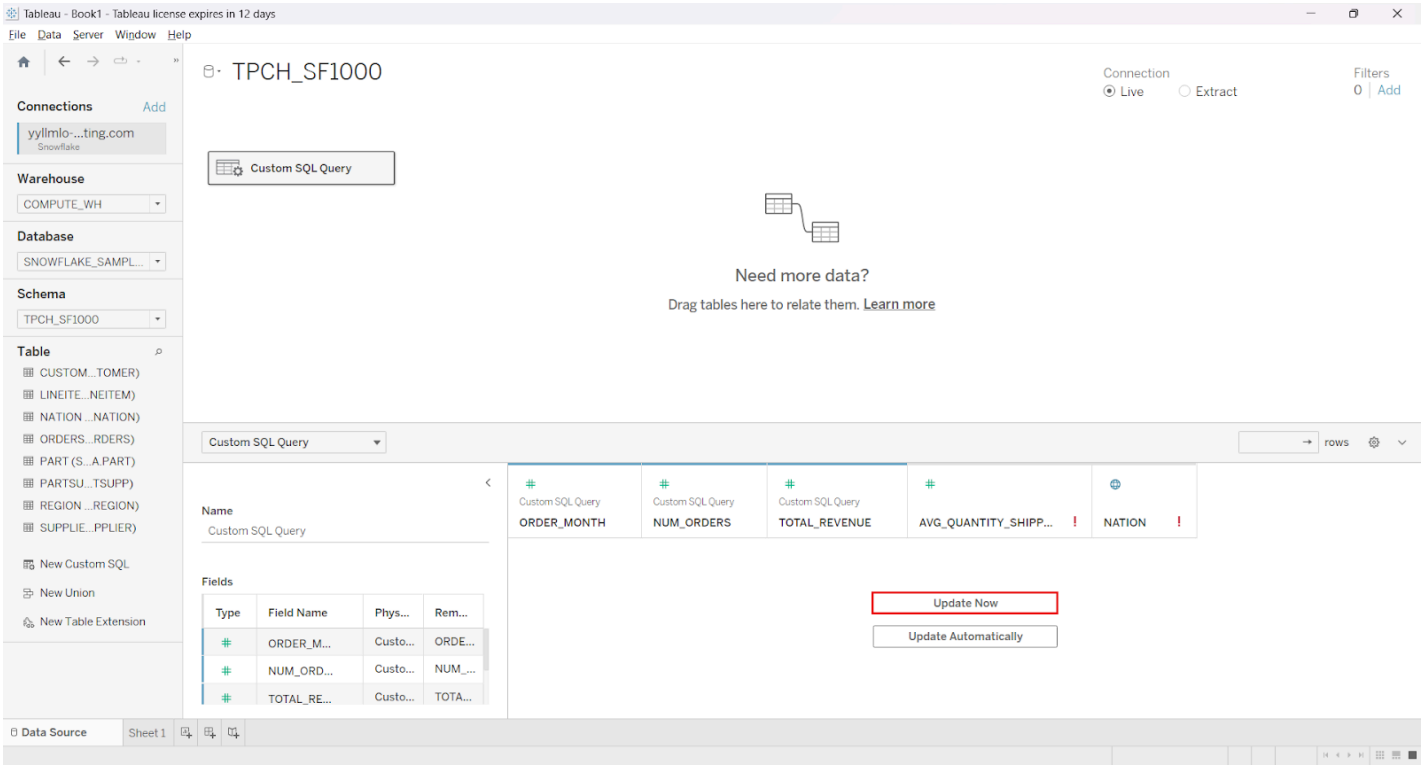


- Let us now edit the custom query in Tableau Desktop and use the new query to fetch the data. Use below query:

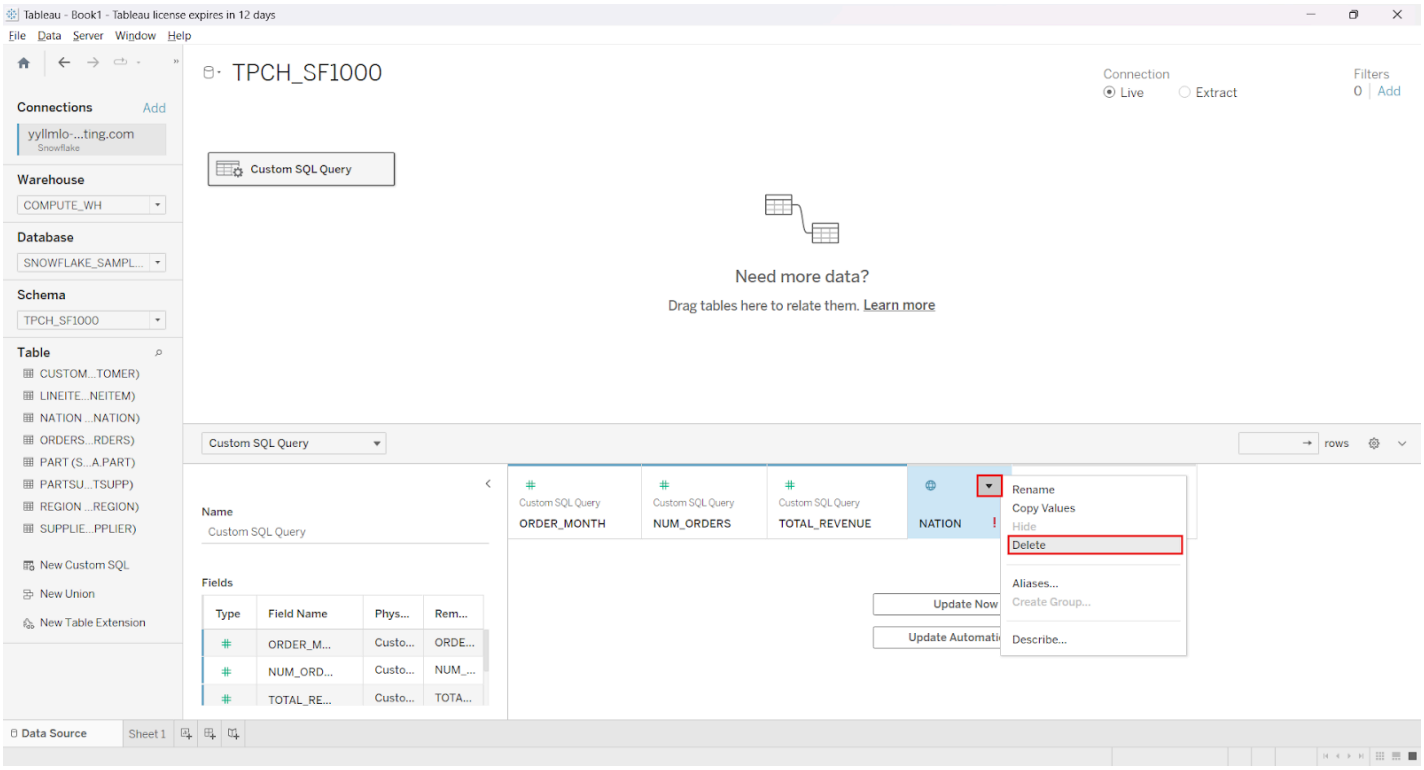
```
SELECT
 (CASE WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 1 THEN 'JANUARY'
  WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 2 THEN 'FEBRUARY'
  WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 3 THEN 'MARCH'
  WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 4 THEN 'APRIL'
  WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 5 THEN 'MAY'
  WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 6 THEN 'JUNE'
  WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 7 THEN 'JULY'
  WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 8 THEN 'AUGUST'
  WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 9 THEN 'SEPTEMBER'
  WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 10 THEN 'OCTOBER'
  WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 11 THEN 'NOVEMBER'
  WHEN EXTRACT(MONTH FROM o.O_ORDERDATE) = 12 THEN 'DECEMBER'
  END) as order_month,
 COUNT(DISTINCT o.O_ORDERKEY) as num_orders,
 SUM(l.L_EXTENDEDPRICE * (1 - l.L_DISCOUNT)) as total_revenue
FROM
 snowflake_sample_data.tpch_sf1000.orders o
JOIN
 snowflake_sample_data.tpch_sf1000.lineitem l
ON
 o.O_ORDERKEY = l.L_ORDERKEY
WHERE
 EXTRACT(YEAR FROM o.O_ORDERDATE) = '1997'
GROUP BY
 order_month
ORDER BY
 order_month
```



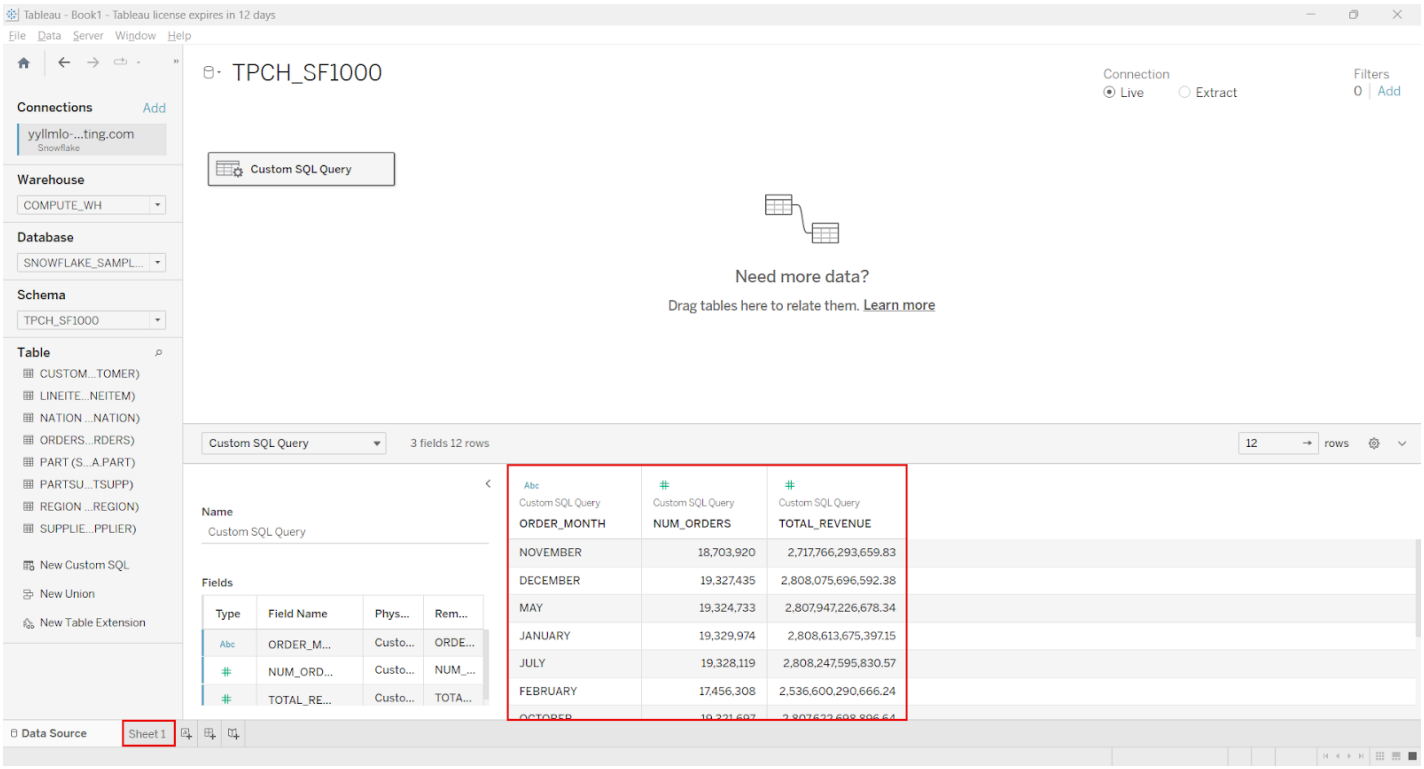
Now update the data in Tableau.



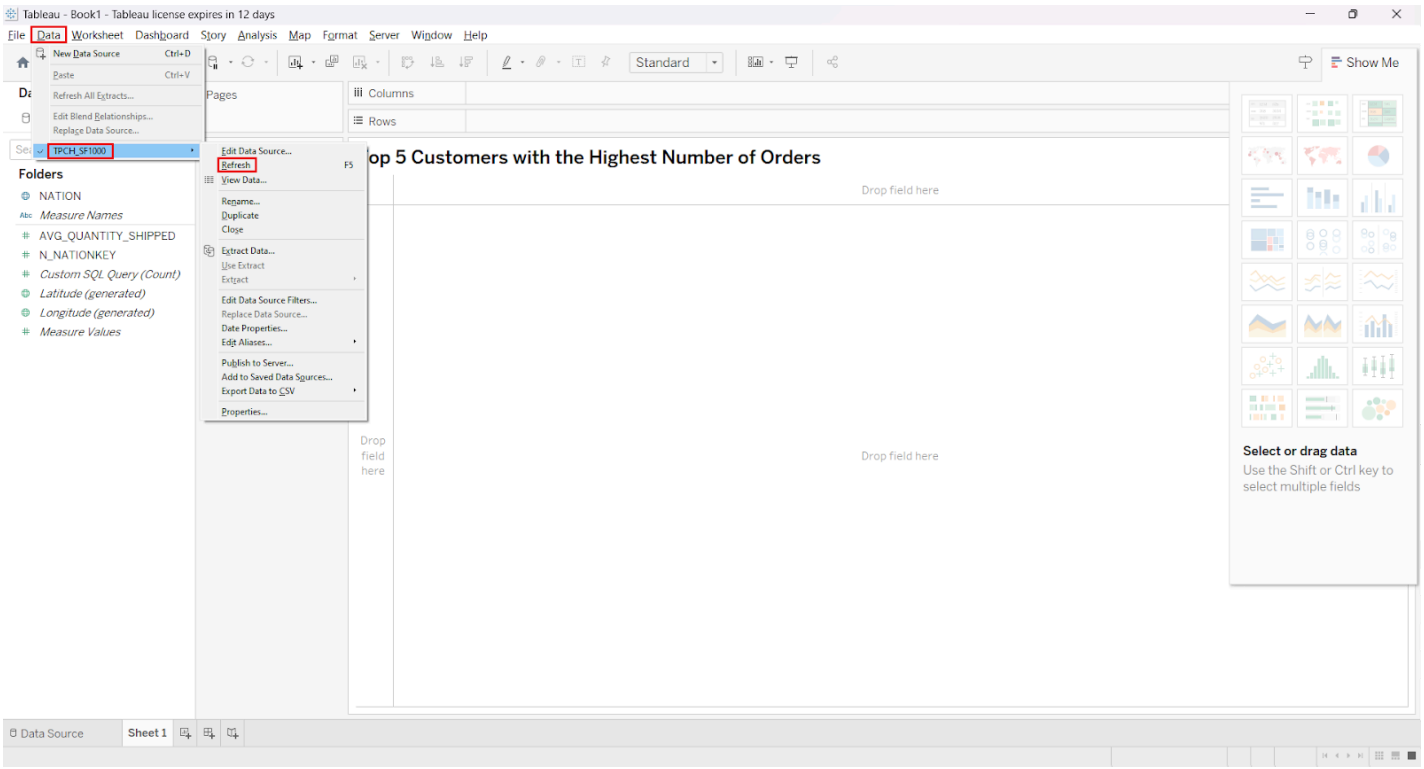
If you want to remove certain fields you can hover over those fields and click on down arrow and click on Delete.



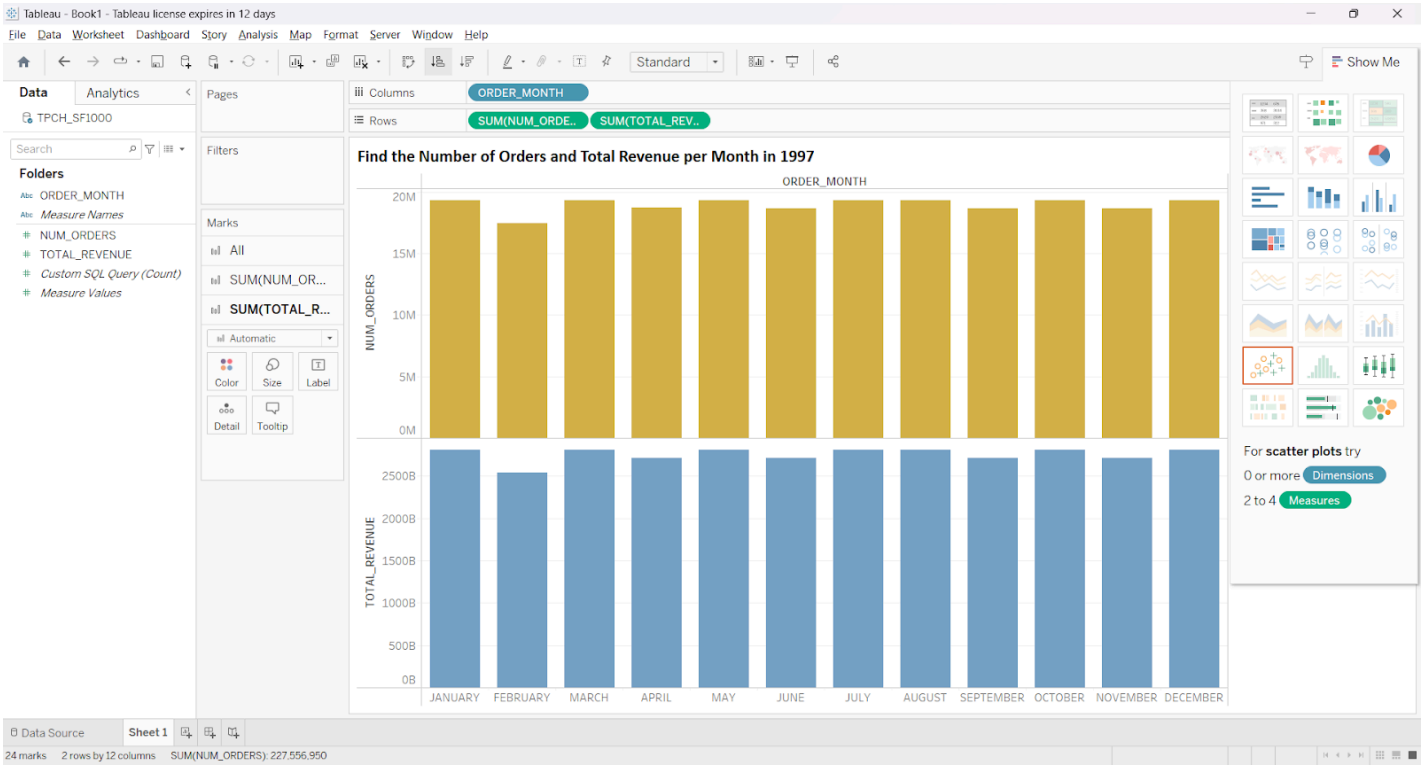
- After the update is done go to sheet1.



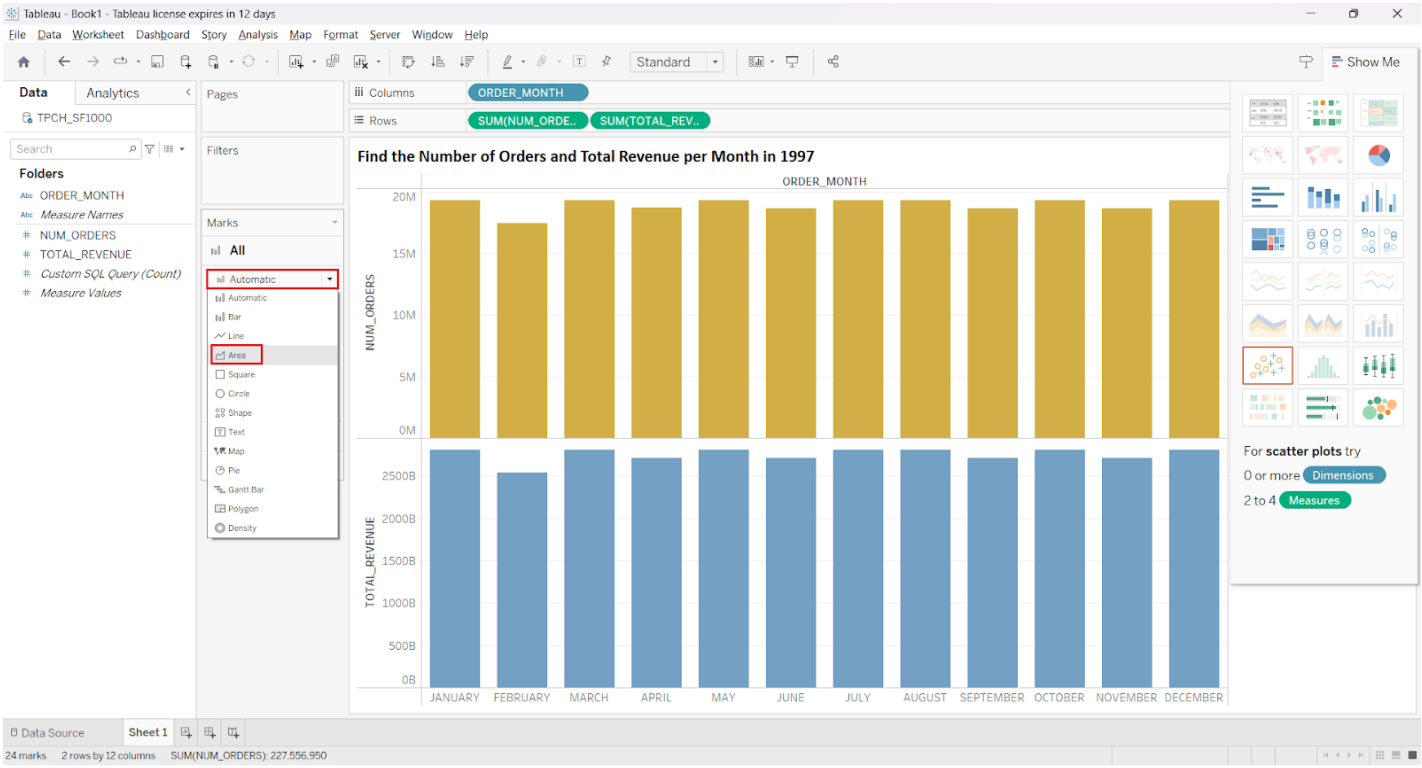
Then click on Data menu then hover over your schema and then click on Refresh data.



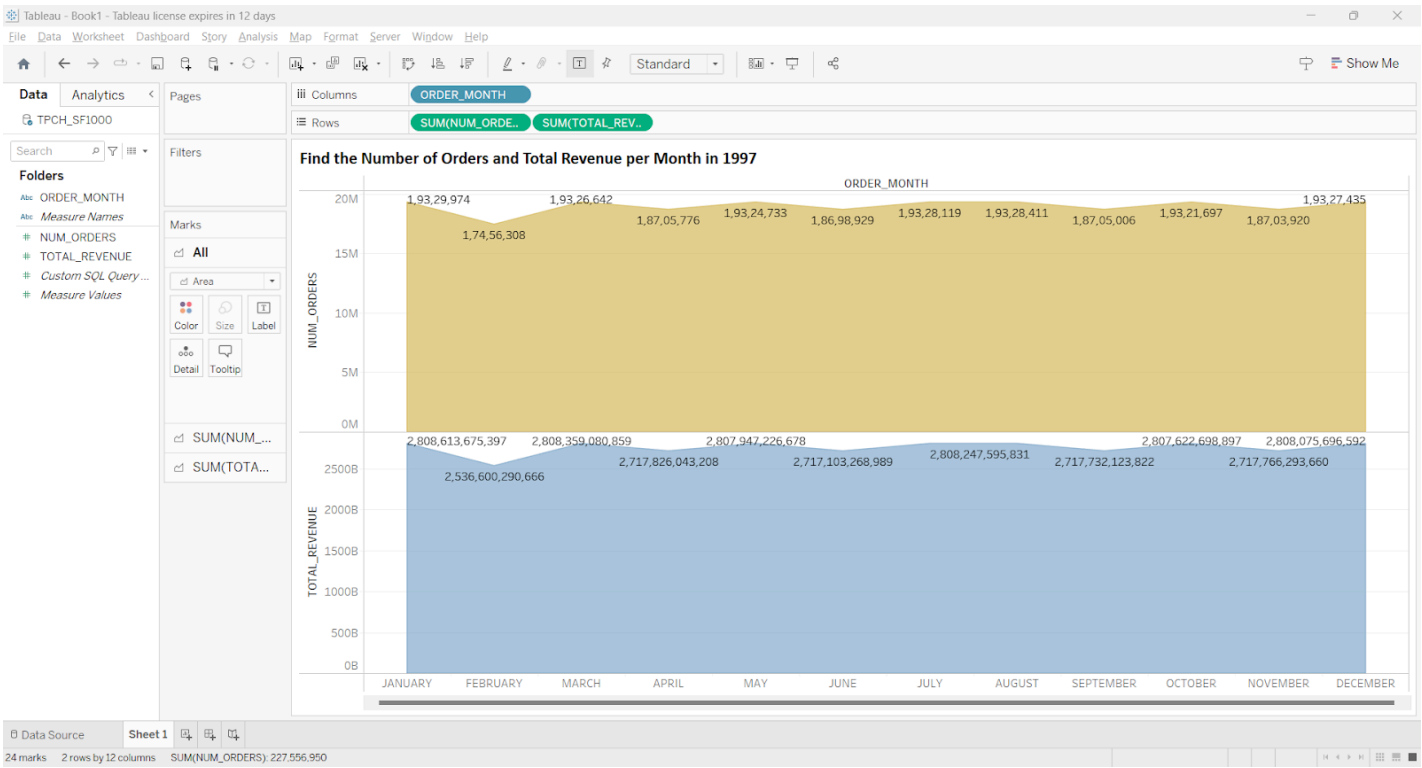
- Now you can see the updated dimensions and measures. Now edit the title. And we will plot ORDER_MONTH vs TOTAQL_REVENUE & NUM_ORDERS. After you have plotted the plot will look like below. You may choose to give different colours and add more labels.



Automatically the best representation is considered though you can choose nay representation of your choice. Let us use a line graph in this case. Click on Automatic under marks pane and the select Area.



Now the graph looks like below.



Else you can choose from available graphs on the right hand section under Show Me .

YOU HAVE SUCCESSFULLY CREATED VISUALIZATION FOR YOUR DATA IN SNOWFLAKE